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AI Turf “War”: An Old Man’s Defence of Good Old- Fashioned AI

When I began my graduate studies in Artificial Intelligence half a decade ago, AI was still popular but far from being in the limelight as it is today. Seminal research work, which would lead to the amazingly realistic image generation models we see today, was yet to be published. GPT-1 was just a year old. The idea of feeding large amounts of publicly available data to large transformer-based models - the beating heart of any large language model you can start using today - was just getting popular in academia and industry. ML and AI were far from being dormant though. It was a space humming with activity around what is now informally referred to as “Traditional AI”.

You might be excused for thinking November 2022 was when the world took note of AI. The ChatGPT moment, albeit a very important one in the meteoric launch of AI into the permanent mainstream fixture of today, was far from being representative of the myriad ways in which Machine Learning (ML) was integrated into consumer and business products. Generative AI (genAI) was promising and did exist then — with Generative Adversarial Networks (GANs) as the star player (check out

<https://www.whichfaceisreal.com/> to see an artifact from that era) — but with far and few broad commercial applications. The current generation of genAI solutions is indeed a world apart from the past but what happened to traditional AI? Where is it now? And where will it be in the future?

Before I make claims about where we are and where we might go in the future, I think it would be best to clarify what traditional AI is generally considered to be. Traditional AI’s primary purpose is discriminative (as opposed to generative) in nature. It doesn’t “create” anything new. It is less concerned with filling in the next word or pixel and more concerned with extracting useful features of your input data to make very task-specific predictions.

Think about the algorithm your bank uses to determine whether that 50K transaction made on your credit card a minute ago in Eastern Europe is fraudulent or not. Or the one that Netflix/Instagram/Tiktok etc uses to recommend the next piece of media to consume, which will most likely increase your sleepless binge session by a few more hours. Both AIs are “special-purpose” (i.e., Netflix’s trained recommendation engine is unlikely to work well/at all in another

domain like fraud detection). Both AIs take in input data — your transaction/viewing history, broader variables like the behavior of others etc — and transform them into specific predictions (fraud/not, suggest/not).

Traditional AI use cases extend beyond fraud detection or recommendation systems. Deep neural networks that use a certain data transformation technique called a convolution (think of it as smearing two things together to find out overlaps between the two), have been the workhorse for a wide variety of image-related tasks like image classification, object detection, image segmentation etc for a long time. For instance, facial recognition systems currently in production for Big Brother-style surveillance or relatively more benign applications like biometrics, leverage some form of convolutional networks. Many models used for fortune telling aka making predictions about the future aka time series forecasting would also fall under Traditional AI. Think of these as what a clothing manufacturer would use to predict not only the demand for their sturdy jeans next summer but also for the raw materials they would need to purchase in advance to fulfill that future demand. I can go on and on.